

Supporting Information for: Scalable Simulation of Non-Markovian Energy Transfer via Process Tensor Hierarchy of Pure States with Spectrally Bundled Dissipators

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1 FMO Hamiltonian and Spectral Density Parameters

While the PT-HOPS/SBD formalism and 12-test validation benchmarking are documented comprehensively in the main text, this Supporting Information explicitly provides the 7-site bacteriochlorophyll-*a* matrices utilized in our numerical models (from Ref. [1]).

1.1 Site Energies

Table 1: FMO Site Energies (cm^{-1})

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

1.2 Electronic Couplings

The inter-chromophore electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \quad (1)$$

1.3 Spectral Density Parameters

References

- [1] Adolphs, J.; Renger, T. Biophys. J. **2006**, *91*, 2778–2797.

Table 2: **Spectral Density Parameters**

Parameter	Value
Drude–Lorentz reorganization (λ_{DL})	35 cm^{-1}
Drude–Lorentz cutoff (γ_{DL})	50 cm^{-1}
Vibronic mode 1 (ω_1)	150 cm^{-1}
Vibronic mode 2 (ω_2)	200 cm^{-1}
Vibronic mode 3 (ω_3)	575 cm^{-1}
Vibronic mode 4 (ω_4)	$1\,185 \text{ cm}^{-1}$
Huang–Rhys S_1	0.05
Huang–Rhys S_2	0.02
Huang–Rhys S_3	0.01
Huang–Rhys S_4	0.005